

Supplementary Table 1. Two-way ANOVA, site 83 in wild-type backgrounds

<i>Source of Variation</i>	<i>SS</i>	<i>d.f.</i>	<i>MS</i>	<i>F</i>	<i>p-level</i>	<i>F crit</i>	<i>Omega Sqr.</i>
<i>Factor #1 (Species)</i>	2.05013	1	2.05013	0.83931	0.38638	5.31766	0.
<i>Factor #2 (site 83)</i>	167.25333	1	167.25333	68.47255	0.00003	5.31766	0.65788
<i>Factor #1 + #2 (Species x site 83)</i>	59.22963	1	59.22963	24.24827	0.00116	5.31766	0.22668
<i>Within Groups</i>	19.54107	8	2.44263				
<i>Total</i>	248.07417	11	22.5522				
<i>Omega squared for combined effect</i>	0.883						

Post-hoc comparisons

Comparisons among groups of factor 2 (site 83) within each factor 1 (Species) level

Factor 1 (Species) = Bos taurus

Tukey HSD

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
D vs N	-11.91	13.19907	0.00018	Yes

Factor 1 (Species) = Orcinus orca

Tukey HSD

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
D vs N	-3.02333	3.35056	0.05535	No

Comparisons among groups of factor 1 (Species) within each factor 2 (site 83) level

Factor 2 (site 83) = D

Tukey HSD

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
Bos taurus vs Orcinus orca	-5.27	5.84039	0.00341	Yes

Factor 2 (site 83) = N

Tukey HSD

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
Bos taurus vs Orcinus orca	3.61667	4.00811	0.02203	Yes

Supplementary Table 2. Two-way ANOVA, site 83 in S292/A299 backgrounds

<i>Source of Variation</i>	<i>SS</i>	<i>d.f.</i>	<i>MS</i>	<i>F</i>	<i>p-level</i>	<i>F crit</i>	<i>Omega Sqr.</i>
<i>Factor #1 (Species)</i>	0.2312	1	0.2312	0.84126	0.41093	7.70865	0.
<i>Factor #2 (site 83)</i>	105.85125	1	105.85125	385.15874	0.00004	7.70865	0.9768
<i>Factor #1 + #2 (Species x site 83)</i>	0.6272	1	0.6272	2.28218	0.20539	7.70865	0.00326
<i>Within Groups</i>	1.0993	4	0.27483				
<i>Total</i>	107.80895	7	15.40128				
<i>Omega squared for combined effect</i>	0.97966						

Post-hoc comparisons

Comparisons among groups of factor 2 (site 83) within each factor 1 (Species) level

Factor 1 (Species) = Bos taurus

Tukey HSD

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
D vs N	-7.835	21.13615	0.00032	Yes

Factor 1 (Species) = Orcinus orca

Tukey HSD

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
D vs N	-6.715	18.11477	0.0004	Yes

Comparisons among groups of factor 1 (Species) within each factor 2 (site 83) level

Factor 2 (site 83) = D

Tukey HSD

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
Bos taurus vs Orcinus orca	-0.9	2.42789	0.16128	No

Factor 2 (site 83) = N

Tukey HSD

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
Bos taurus vs Orcinus orca	0.22	0.59348	0.69624	No

Supplementary Table 3. Two-way ANOVA, site 83 in A292/S299 backgrounds

<i>Source of Variation</i>	<i>SS</i>	<i>d.f.</i>	<i>MS</i>	<i>F</i>	<i>p-level</i>	<i>F crit</i>	<i>Omega Sqr.</i>
<i>Factor #1 (Species)</i>	5.13601	1	5.13601	1.32708	0.31349	7.70865	0.005
<i>Factor #2 (site 83)</i>	154.96801	1	154.96801	40.04173	0.00319	7.70865	0.59645
<i>Factor #1 + #2 (Species x site 83)</i>	73.87201	1	73.87201	19.08757	0.01198	7.70865	0.27633
<i>Within Groups</i>	15.48065	4	3.87016				
<i>Total</i>	249.45669	7	35.63667				
<i>Omega squared for combined effect</i>	0.87778						

Post-hoc comparisons**Comparisons among groups of factor 2 (site 83) within each factor 1 (Species) level****Factor 1 (Species) = Bos taurus****Tukey HSD**

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
D vs N	-14.88	10.69679	0.0018	Yes

Factor 1 (Species) = Orcinus orca**Tukey HSD**

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
D vs N	-2.725	1.95892	0.23808	No

Comparisons among groups of factor 1 (Species) within each factor 2 (site 83) level**Factor 2 (site 83) = D****Tukey HSD**

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
Bos taurus vs Orcinus orca	-4.475	3.21694	0.08537	No

Factor 2 (site 83) = N**Tukey HSD**

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
Bos taurus vs Orcinus orca	7.68	5.52092	0.01762	Yes

Supplementary Table 4. Two-way ANOVA, site 292 in wild-type backgrounds

<i>Source of Variation</i>	<i>SS</i>	<i>d.f.</i>	<i>MS</i>	<i>F</i>	<i>p-level</i>	<i>F crit</i>	<i>Omega Sqr.</i>
<i>Factor #1 (Species)</i>	249.76125	1	249.76125	195.29762	0.00015	7.70865	0.90555
<i>Factor #2 (Site 292)</i>	15.07005	1	15.07005	11.78383	0.02647	7.70865	0.05026
<i>Factor #1 + #2 (Species x Site 292)</i>	3.1752	1	3.1752	2.48281	0.19022	7.70865	0.00691
<i>Within Groups</i>	5.1155	4	1.27887				
<i>Total</i>	273.122	7	39.01743				
<i>Omega squared for combined effect</i>	0.96272						

Post-hoc comparisons

Comparisons among groups of factor 2 (Site 292) within each factor 1 (Species) level

Factor 1 (Species) = Bos taurus

Tukey HSD

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
A vs S	4.005	5.00845	0.02412	Yes

Factor 1 (Species) = Orcinus orca

Tukey HSD

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
A vs S	1.485	1.85707	0.25926	No

Comparisons among groups of factor 1 (Species) within each factor 2 (Site 292) level

Factor 2 (Site 292) = A

Tukey HSD

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
Bos taurus vs Orcinus orca	-9.915	12.3992	0.00111	Yes

Factor 2 (Site 292) = S

Tukey HSD

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
Bos taurus vs Orcinus orca	-12.435	15.55059	0.00056	Yes

Supplementary Table 5. Two-way ANOVA, site 299 in wild-type backgrounds

<i>Source of Variation</i>	<i>SS</i>	<i>d.f.</i>	<i>MS</i>	<i>F</i>	<i>p-level</i>	<i>F crit</i>	<i>Omega Sqr.</i>
<i>Factor #1 (Species)</i>	43.47781	1	43.47781	181.28029	0.00018	7.70865	0.57436
<i>Factor #2 (Site 299)</i>	28.38811	1	28.38811	118.36394	0.00041	7.70865	0.37391
<i>Factor #1 + #2 (Species x Site 299)</i>	2.21551	1	2.21551	9.23756	0.03843	7.70865	0.02624
<i>Within Groups</i>	0.95935	4	0.23984				
<i>Total</i>	75.04079	7	10.72011				
<i>Omega squared for combined effect</i>	0.97451						

Post-hoc comparisons

Comparisons among groups of factor 2 (Site 299) within each factor 1 (Species) level

Factor 1 (Species) = Bos taurus

Tukey HSD

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
A vs S	-2.715	7.84018	0.00531	Yes

Factor 1 (Species) = Orcinus orca

Tukey HSD

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
A vs S	-4.82	13.91885	0.00078	Yes

Comparisons among groups of factor 1 (Species) within each factor 2 (Site 299) level

Factor 2 (Site 299) = A

Tukey HSD

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
Bos taurus vs Orcinus orca	-3.61	10.4247	0.00198	Yes

Factor 2 (Site 299) = S

Tukey HSD

<i>Groups</i>	<i>Difference</i>	<i>Test Statistic</i>	<i>p-level</i>	<i>Significant</i>
Bos taurus vs Orcinus orca	-5.715	16.50337	0.00048	Yes

Supplementary Table 6. Likelihood ratio tests for random-sites models (PAML) of the rhodopsin species tree

Model	np	ln <i>L</i>	κ	Parameters ^a			Null	LRT	df	p	AIC
				ω_0/p	ω_1/q	ω_2/ω_p					
M0	63	-4704.03	4.44	0.07							9534.06
M1a	64	-4572.35	4.66	0.03 (91.0%)	1 (9.0%)						9272.7
M2a	66	-4572.35	4.66	0.03 (91.0%)	1 (7.0%)	1.00 (2.0%)	M1a	0	2	1.000	9276.7
M3	67	-4543.37	4.55	0.00 (70.0%)	0.11 (21.0%)	0.61 (9%)	M0	321.3	4	0.000	9220.74
M7	64	-4544.21	4.55	0.09	1.01						9216.42
M8a	65	-4543.53	4.56	0.1	1.33	1.0 (1.0%)					9217.06
M8	66	-4543.27	4.57	0.1	1.26	1.39 (1.0%)	M7	1.9	2	0.387	9218.54
							M8a	0.5	1	0.480	

Note: np, number of parameters; ln *L*, ln likelihood; κ , transition/transversion ratio; df, degrees of freedom. ^aFor models M0-M3, the ω values for each site class (ω_0 - ω_2) are shown with their proportions in parentheses. For models M7-M8, *p* and *q* describe the shape of the beta distribution, and ω_p refers to the positively selected site class (with proportion in parentheses) for models M8 and M8a (where it is constrained to one).

Supplementary Table 7. Sources for rhodopsin sequences used in ancestral sequence reconstruction

Common name	Binomen	Accession number
African elephant	<i>Loxodonta africana</i>	AY686752.1
Human	<i>Homo sapiens</i>	NM_000539.3
Domestic cat	<i>Felis catus</i>	NM_001009242.1
Bactrian camel	<i>Camelus bactrianus</i>	XM_010953086.1
Wild boar	<i>Sus scrofa</i>	NM_214221.1
Sheep	<i>Ovis aries</i>	XM_004018534.3
Water buffalo	<i>Bubalis bubalis</i>	XM_006078900.1
Plains bison	<i>Bison bison</i>	XM_010862448.1
Cattle	<i>Bos taurus</i>	NM_001014890.1
Hippo	<i>Hippopotamus amphibius</i>	KC676928.1
Bowhead whale	<i>Balaena mysticetus</i>	KC676921.1
N. Atlantic right whale	<i>Eubalaena glacialis</i>	JQ730751.1
Pygmy right whale	<i>Caperea marginata</i>	KC676926.1
N. Atlantic minke whale	<i>Balaenoptera acutorostrata acutorostrata</i>	KC676922.1
Blue whale	<i>Balaenoptera musculus</i>	KC676923.1
Fin whale	<i>Balaenoptera physalus</i>	KC676924.1
Sperm whale	<i>Physeter macrocephalus</i>	XM_007126220.1
South-Asian river dolphin	<i>Platanista minor</i>	KC676936.1
Sowerby's beaked whale	<i>Mesoplodon bidens</i>	AF055316.1
Baird's beaked whale	<i>Berardius bairdii</i>	KC676925.1
Cuvier's beaked whale	<i>Ziphius cavirostris</i>	KC676938.1
Yangtze river dolphin	<i>Lipotes vexillifer</i>	XM_007461564.1
Franciscana	<i>Pontoporia blainvillei</i>	KC676937.1
Amazon river dolphin	<i>Inia geoffrensis</i>	KC676929.1
Beluga	<i>Delphinapterus leucas</i>	KC676927.1
Finless porpoise	<i>Neophocaena phocaenoides</i>	KC676932.1
Harbour porpoise	<i>Phocoena phocoena</i>	KC676933.1
Dall's porpoise	<i>Phocoenoides dalli</i>	KC676934.1
Killer whale	<i>Orcinus orca</i>	XM_004284305.1
Bottlenose dolphin	<i>Tursiops truncatus</i>	AF055456.1
Pilot whale	<i>Globicephala melas</i>	AF055315.1
Common dolphin	<i>Delphinus delphis</i>	AF055314.1

Supplementary Text: Statistical analyses

To determine the effect of site 83 on retinal release and how it interacts with sites 292 and 299, we conducted a series of two-way analyses of variance (ANOVAs) that compared the D83 vs. N83 residues in the killer whale and bovine rhodopsin backgrounds. Comparisons between the species were made in the wild-type backgrounds, and two backgrounds where sites 292 and 299 were mutated to have the same residue in each species (S292/A299, A292/S299). To ensure the ANOVAs were balanced (each factor with the same sample size, n , where n is the number of rhodopsin elutions), where necessary we randomly subsampled from retinal release $t_{1/2}$ measurements. Measures of effect size were calculated using Hays omega-squared correlation coefficient. In each ANOVA, all factors had non-significant ($p > 0.05$) differences in variance (Levene's tests) and residuals were normally distributed. To correct for multiple tests, post-hoc pairwise comparisons between ANOVA groups were assessed using Tukey's statistic.